

SMART UTM — a tensile rig that runs, watches and stops its own tests

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The camera measures what a certified extensometer measures.

20 smart features and 6 closed-loop protocols on a desktop rig. On the same frames, the rig's DIC agrees with a Zwick/Roell XT-205 video extensometer to 0.07 % in scale and 0.19 % in fracture strain — and PLA, PETG and TPU land inside their published bands.

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smart features
rig-validated (+2 offline)

0.99928

scale vs XT-205 on the
same frames, R² 0.9999964

±1 %

yield and fracture strain
vs the laboratory

12 µε

DIC noise floor, 10 s —
reference instrument 45 µε

1.78× / 1.78×

noise vs pixel span
predicted / measured

3

polymers inside their
published bands

THE MACHINE

The rig — built in-house, every control loop closed in software

Two steppers (1.85 Nm, 20:1) drive 5 mm lead screws against a fixed base. An ANYLOAD 3 t cell measures force; a Basler camera measures strain optically at 20.8 px/mm from two markers on the 80 mm gauge. The ESP32 runs STEP/DIR/ENABLE only and knows nothing of test protocols — every protocol, guard and measurement here lives in the PyQt6 app, closing the loop at ~11 Hz.

WHY AUTOMATE A TENSILE RIG

A conventional pull gives ONE modulus, ONE yield, ONE strength — and a human presses every button. Crosshead travel is not specimen strain, so the machine reports its own compliance as material behaviour. A destructive test is one irreversible shot. And nothing watches the run, so a stall, a lost marker or a fracture are all noticed late, by a person.

HOW IT WORKS

One control engine

All six protocols are one function called on every load sample (~11 Hz) with load, position and time-matched DIC strain in hand. A protocol only returns a speed — it cannot bypass the shared safety net.

Strain from the specimen

A camera tracks two markers; engineering strain is (L–L₀)/L₀ at 20.8 px/mm. The centroid resolves 0.1 px steps at ±0.02 px.

Simulated before hardware

Every policy is replayed against a spring plant and recorded CSVs — 9/9 checks pass before a specimen is risked.

Four safety layers

Load-collapse detector · stall guard (< 0.05 mm / 6 s under load) · 4.5 kN and 30 mm backstops · dead-DIC guard (freeze at 0.2 s, halt at 1.0 s). Guards are phase-aware, so the stall guard stays silent through an intentional dwell.

WHAT THE TESTS SHOWED

Validated against a certified video extensometer — on its own frames

The laboratory's XT-205 video was re-measured by the rig's post-processor: 3153 frames, scale 0.99928, R² 0.9999964, fracture strain 5.21 vs 5.22 %, and a constant offset of -55 µε — 0.069 px on a 1255 px reference — that vanishes from every slope. On separate specimens the two machines agree within 1 % on yield and fracture strain; the -7.7 % UTS gap is pull rate (2.21× the laboratory's), grip seating and batch, not the strain channel.

Three polymers, three characters — each inside its published band

PLA S25: UTS 46.2 MPa, E 3.17 GPa, ε_f 4.1 %. PETG S30: 41.8 MPa, 1.81 GPa, 7.8 %. TPU S37: 2.60 MPa at 18.9 % strain, E 0.024 GPa, unbroken at the travel limit. Two decades of modulus with the same two markers and the same code, in the expected order — the rig measures the material, not itself.

WHAT IT MEANS

Same frames, same answer → the software is a measurement

Given the XT-205's own frames, the rig's algorithm reproduces the certified record to 0.07 % in scale. Whatever still differs between the two machines is the machine or the specimen, not the strain channel.

The noise is centroid noise → design with pixels

Floor 11.9 µε over 10 s, 26 µε over 900 s, below the reference instrument's 45 µε. It scales as 1/Px₀: 1675 vs 939 px across the gauge predicted 1.78× and measured 1.78×. More pixels across the gauge is the only lever, and it needs no code.

The crosshead is not the specimen → an extensometer is not optional

Only 19–29 % of early crosshead motion reaches the gauge; a crosshead-based curve overstates strain, and understates modulus, by three to five times.

A fixed threshold beats Otsu → material presets, not auto-thresholds

Otsu is set by the specimen body, not the markers (painting them out moves it ≤ 3 grey levels). On translucent PETG it found both markers on 48.5 % of frames; the fixed cut on 99.5 %.

Material, not machine → the rig can rank materials

A matched 50 % pair repeats to 0.2 % in strength (19.87 / 19.91 MPa); the 50 % infill knockdown is 2.4 in both strength and modulus, against the 2.0 that volume fraction alone would give; fractures do not cluster at the markers.

Batch is a variable → control the spool

The same PLA spool two months later printed parts with 1.7–2.5× the fracture strain; the literature gives –8.5 % strength after 24 h of ambient exposure. Specimens printed weeks apart are not the same material.

The cost is the thinking, not the pulling

The whole project: 3.2 kg CO₂e, 83 g per recorded test; the rig itself is 13 g per test and the desk computer outweighs it 239×.

LIMITATIONS

Poisson's ratio is not measurable today

The gauge is too narrow: the elastic width change is sub-pixel at 20.9 px/mm. Every stress here is ENGINEERING (F/A₀); true stress is an estimate. Future work FW1: a second camera or a lens change.

The UTS gap to the laboratory is explained, not yet separated

–7.7 % on one specimen each side. Pull rate (2.21×), grip seating and batch are the candidates; one pull at the laboratory's 2 mm/min and the laboratory's load channel would split them.

THE SMART FEATURES, SF1 TO SF20

built & rig-validated (18)

built, verified offline (2)

Live DIC health HUD

SF1

A green/amber/red badge shows marker lock, tracking % and jitter live — so a dropout is seen during the test, not found later in the CSV.

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markers, live at ~2 Hz

Prepare specimen

SF2

One button tares position, force and DIC and clears the plots. It tares DIC only at a green 2/2 lock, so a bad mount cannot be zeroed in.

4 → 1

operator actions

Recipes / settings

SF3

Saves and restores an entire setup, including all six protocols' parameters — so repeats are identical by construction, not by memory.

1 click

restores the whole setup

One-click report

SF4

A one-page PDF and every plot, recomputed from the CSV by the same library the app runs live.

1 click

→ PDF + every graph

Auto-stop at fracture

SF5

A load-collapse detector halts the motor the moment the specimen breaks. Caught S16 at 3375 N, silent through the ductile draw before it.

1.2 s

collapse → motor halt

Constant-strain-rate testing

SF6

Closed loop on the specimen's own de/dt: the crosshead adapted itself 0.10 → 0.05 mm/s, fast in the elastic region, slow through necking.

0.05 %/s

held: 0.051 vs a 0.050 %/s target

Stall guard

SF7

Halts a motor that has stopped moving under load. It proved the T7 failure was a 2.6 kN stall, not a fracture — the specimen came off intact.

0.05 mm

trip, and rise above 200 N

Release load

SF8

Taring at preload makes the readout say 0 N while 300 N is still on the specimen. This drives past that, down to the true zero.

≤ 5 N

residual, at 0.20 mm/s

Six closed-loop protocols

SF9

cyclic · staircase · relaxation · creep · staircase→FRACTURE · progressive-cyclic→FRACTURE. One engine, one safety net.

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rig-validated

Auto-preload

SF10

Approaches on a 0.2 → 0.1 → 0.02 mm/s schedule and stops 3 % high, because PLA relaxes about 2 % while held.

1.03 ×

target, pre-paying relaxation

Auto-metadata link

SF11

Every save links the CSV, the captured video and the registry row by name and proposes the per-specimen folder — nothing a run produces can be orphaned.

3 links

CSV · capture · registry, on save

DIC auto-calibrate

SF12

Exposure and threshold are swept on the live camera and scored on marker count, contrast and centroid jitter; the winner is applied with the score shown.

1 click

exposure + threshold, scored

Guided wizard

SF13

Connect → camera → data → mode → auto-calibrate → preload → Px₀ → prepare → run → save. Each step refuses to advance until its check passes.

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steps, on a real run

Test registry

SF14

Each run indexed with whatever its loading path supports — UTS, ε_f, E, σ_y, toughness — plus the force anchor that turns a tared reading into a true load.

36 runs

auto-indexed, 30 specimens

Dead-DIC guard + hard backstops

SF15

A controller ramping blind on a stale camera reading spikes the force. Speed freezes at 0.2 s of frozen strain, halt at 1.0 s; 4.5 kN and 30 mm caps.

3.0×

less overshoot on marker loss (S18 → 175 N)

Live Px₀ overlay

SF16

The frozen reference pair and the live pair are drawn on the feed with a caption whose Δ agrees with the strain readout — the operator sees the measurement, not only a number.

Δ = ε

overlay agrees with readout

Video + image capture

SF17

PNG stills plus three AVI views (raw, contrast-boosted, adaptive speckle) recorded with the run, at a measured cost that never touches the detection budget.

≈47 µs

per frame, all sinks

DIC post-processing

SF18

Measures strain from ANY recorded video with the rig's own pixel-to-strain rule. On the laboratory's XT-205 footage it reproduced the certified record over 3153 frames.

0.99928

scale vs XT-205, same frames

Noise capture

SF19

Records the instrument at rest and separates offset and drift (correctable) from the residual sd — the uncertainty, which does not subtract. Verified offline; awaits a rig run.

26 µε

900 s floor, drift separated

Manual blob selection

SF20

The operator clicks the two markers on a frozen frame and tracking follows the picks with the shape gates off — keeps measuring where automatic detection is deliberately killed.

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offline tests, S13 frames

Code, raw CSVs, the 393-slide progress diary and the project report: Software/UTM_PyQt6/ and documentation/ · Every figure and number here is regenerated from the rig CSVs and the laboratory records by the documentation/scripts analysis suite — none of it is transcribed by hand. · Layout follows the #evenbetterposter / Generation-2 billboard format: a plain-language headline, with methods and limitations kept as full panels rather than sidebars.